

Project

In this project, you will investigate a scalable data management system of your choice*. The project combines a theoretical study of the selected system with a focused investigation of one or more scalability techniques used by that system.

The project contributes 40% of the final course grade. Since this is a 6 ECTS course, a substantial individual and group effort is expected throughout the semester.

The project is organized into several milestones. Please read the requirements and deadlines carefully.

The use of genAI is not permitted for this project. When genAI use (or any other irregularity) is suspected, Article 118 of the [Exam Regulations](#) will be applied and may result in disciplinary sanctions. Since this project is executed in group, we recommend group members to critically assess each others' contributions and concisely track individual contributions. Suspicions of irregularities by group members should be reported to the teaching team immediately.

Task 1: Topic Selection

Deadline: End of Week 3 (see assignment for precise deadline)

Each student must submit:

- A ranked list of preferred systems (most preferred first). For each system, provide:
 - The name of the system;
 - A brief motivation explaining: why the system is interesting; why it fits within the scope of this course; whether it is primarily an OLTP system, an OLAP system, a streaming system, a graph database, a distributed storage system, or another type of scalable data platform.
- Up to three preferred group members (optional).

Based on these submissions, the teaching team will form groups of three students (occasionally two, if necessary) and assign a system to each group. We will make every reasonable effort to honor both your system and group preferences. However, the teaching team reserves the right to assign alternative systems or group compositions when this is

deemed beneficial for the overall organization of the project.

Task 2: Theoretical Investigation of the System

Deadline: End of Week 5 (see assignment for precise deadline)

Each group studies its assigned system and submits a one-page report (single-sided) that provides a concise overview of the system. The report should address the following questions:

- What is the system and what problem does it solve?
- In which contexts and application domains is it used?
- What competing or alternative technologies exist?
- Which design choices does the system make with respect to:
 - Reliability,
 - Maintainability,
 - Scalability?
- Which techniques are used to achieve scalability?
- Particular attention should be paid to techniques that are distinctive, innovative, or significantly different from those used in competing systems. Examples include Bloom filters, vector clocks, eventual consistency, specialized indexing methods, consensus protocols, partitioning strategies, replication techniques, and so forth.

The report should conclude with a ranked list of scalability-related techniques that the group considers most interesting to investigate further. In addition, groups are encouraged to highlight any other aspects that make the system noteworthy, innovative, or particularly relevant to scalable data management.

Required Appendices (not included in the page limit)

- A bibliography containing all consulted resources (scientific publications, technical documentation, blog posts, experiments performed by the group, etc.);
- A description of the contribution of each group member.

Based on this submission, the teaching team will assign one or more techniques for further investigation in the final project. Whenever possible, these techniques will be selected from the list provided by the group. We may also provide feedback on this proposal.

Task 3: Hands-On Investigation of the technique(s)

Deadline: End of Semester (see assignment for precise deadline)

Every group studies their assigned technique(s). Ideally, this is done in the context of the assigned system, but other systems may be involved if relevant.

The final report combines theoretical analysis with practical experimentation.

The report should contain the following components:

1. System Overview (1 page)

A revised version of the report produced for Task 2. Groups may either reuse the original version or submit an updated version (for example based on received feedback).

2. Technical Report (1 page)

A focused discussion of the technique(s) assigned by the teaching team. The report should explain:

- The fundamental ideas behind the technique;
- Why the technique is relevant in the selected system;
- How the technique contributes to scalability;
- Any trade-offs, limitations, or assumptions associated with the technique.

The emphasis should be on understanding both the technique itself and its role within the selected system(s).

3. Experimental Evaluation (1 page)

The report must report on practical experiments that have been conducted to investigating the assigned technique(s). The experiments should be designed to provide insight into the behavior, advantages, limitations, or performance implications of the technique.

4. Required Appendices

The following appendices are mandatory and do not count towards the page limit:

- Bibliography containing all consulted sources (this of course includes the bibliography of milestone 2);
- Individual contribution statement;
- Reflection on received feedback, explaining how feedback was incorporated or why certain suggestions were not adopted;
- Reproducibility appendix containing:
 - Raw experimental data;

- Configuration information;
- Scripts and code references;
- Any additional material required to reproduce the reported results.

The final report is submitted once per group.

In addition, **each group member must individually submit a genAI declaration** describing any use of generative AI tools during the project.

Task 3B: Peer Feedback (Optional)

deadline: End of week 10 (see assignment for precise deadline) Feedback will be due one week later.

Groups may submit a preliminary version of the final report for feedback. Submitted projects will be distributed for peer review and will also receive feedback from the teaching team.

Feedback should address aspects such as:

- Clarity and precision;
- Technical correctness;
- Depth and completeness of the discussion;
- Quality of the experimental methodology;
- Interpretation of results;
- Overall coherence and structure of the report;
- Questions regarding the operation or design of the system;
- Missing considerations, assumptions, or limitations.

Students are encouraged to provide constructive, actionable feedback.

Task 4: Project Defense

See exam schedule for the precise date.

During the examination period each group will have the chance to defend their project in an oral question-and-answer session.

Every student is expected to have a solid understanding of all aspects of the project, including:

- The selected system;
- The investigated techniques;
- The experimental methodology;
- The reported results and conclusions.

Individual understanding may be assessed during the defense.

Grading Milestones

- 1: Topic Selection - Not graded (mandatory)
- 2: Theoretical investigation - Not graded (mandatory)
- 3: Final Report - Graded: 80% of the project mark (mandatory)
- 3B: Peer Feedback - Not graded (optional)
- 4: Project Defense - Graded: 20% of the project mark (mandatory)

Grade Composition

Separate marks are assigned for the written report and the oral question-and-answer session. In principle, each mark will be assigned to the group (hence with students of a group receiving the same mark).

In case we have indications that contributions of group members are severely misbalanced, we may decide to deviate from this principle and assign different grades taking contributions (for the written report) or answers (for the project defense) into account.

If the defense component receives a mark below 8/20, that mark automatically becomes the final project grade. Otherwise, the final project grade is computed as a weighted sum: 80% Theoretical and Hands-on work + 20% defense.

Late Submission Policy

- **Milestones 1 and 2:** No formal penalty applies. However, late submissions may delay group formation, topic assignment, or feedback. Any resulting consequences remain the responsibility of the student. Do note that these milestones are mandatory to be allowed to partake in Milestone 3.
- **Milestone 3:** Late submissions are not allowed and will result in an ABSENT mark for the entire project, and consequently, an ABSENT mark for the course.
- **Milestone 3B:** No penalty applies. However, students forfeit the opportunity to receive feedback.
- **Milestone 4:** Absence during the project defense results in an ABSENT mark for the entire project and, consequently, an ABSENT mark for the course.

Second Examination Session

In case the project failed, it can be reworked during the second examination session. In that case, only the Final Report (Milestone 4) and the project defense (Milestone 3) apply during

the second examination session. No feedback opportunities will be provided.

Groups are expected to continue working on the **approved** project topic. Requests for a change of topic or group composition may be submitted but require approval from the teaching team. Such requests should be made before the start of the summer recess. Late requests may not be accommodated.